INCH-POUND
MS27785D
w/Amendment 1
4 April 2016
SUPERSEDING
MS27785D
18 April 2011

DETAIL SPECIFICATION SHEET

SWITCH, TOGGLE, TWO POLE, ENVIRONMENTALLY SEALED, INTEGRATED WIRE TERMINALS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-DTL-3950.

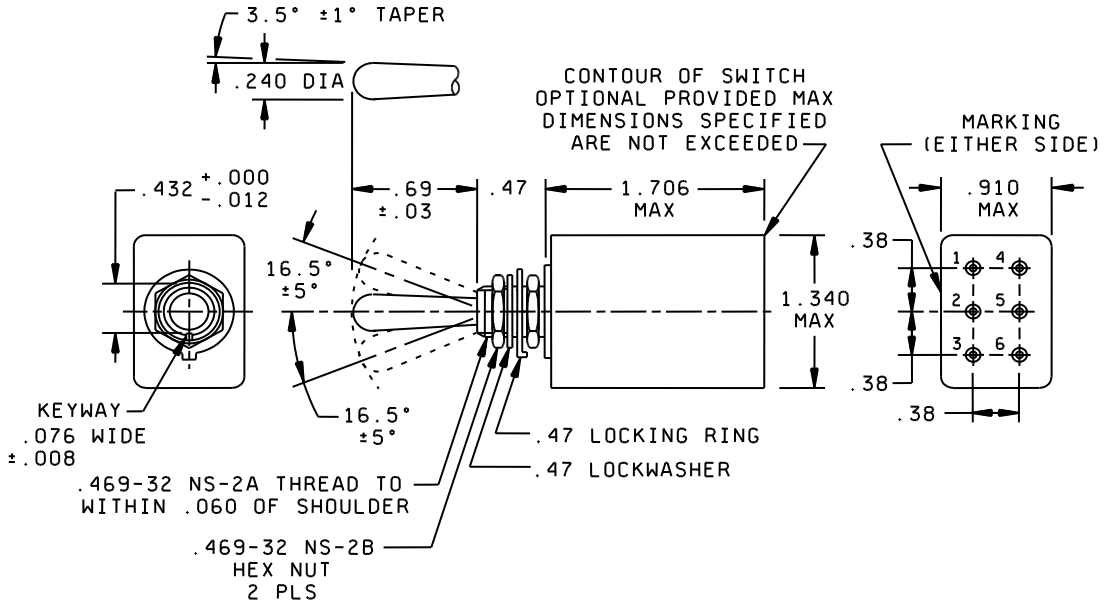


FIGURE1. Dimensions and configuration

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Inches	mm	Inches	mm
.005	0.13	.38	9.7
.008	0.20	.432	10.97
.012	0.30	.469	11.91
.020	0.51	.47	11.9
.03	0.8	.69	17.5
.060	1.52	.910	23.11
.076	1.93	1.340	34.04
.240	6.10	1.706	43.33

NOTES:
1. Dimensions are inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .020 (0.51 mm) for two place decimals and ± .005 (0.13 mm) for three place decimals.
4. For hardware detail specifications see appendix of MIL-DTL-3950.
5. Contact installing/ removal tool in accordance with M81969/14-02.
6. Grommet sealing plug, in accordance with MS27488-A20.
7. Terminal shall adequately accept a wire contact within dimensional limits of SAE-AS39029/1-101.
8. In the event of a conflict between the text of this standard and the reference cited herein, the text of this standard shall take precedence.

FIGURE1. Dimensions and configuration- Continued

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REQUIREMENTS

The Sealing grommet shall seal on a smooth wire insulation of .040 (1.02 mm) to .083 (2.11mm) diameters

The terminal end of switch shall be color coded red to indicate contact size

Direction of internal mechanism movement is opposite to the direction of the toggle movement

Sealing plugs may be used in nonfunctional grommet holes.

Maximum weight is .135 pound.

Electrical rating: See table I.

TABLE I Detail Requirements

MS Part Number	Circuit with toggle lever in			Current capacity (amperes) 28 volts dc			Current capacity (amperes) 115 volts, 60 and 400 hertz ac		
	Keying Side	Center	Opposite keying side	Lamp-Load circuit	Resistive circuit	Inductive circuit	Lamp-Load circuit	Resistive circuit	Inductive circuit
MS27785-1	1-2 ON 4-5	1-2 ON 5-6	2-3 ON 5-6	7	7.5	7.5	4	7.5	7.5
MS27785-2	MOM. 1-2 ON 4-5	1-2 ON 5-6	2-3 ON 5-6	5	7.5	7.5	2	7.5	7.5
MS27785-3	MOM. 1-2 ON 4-5	1-2 ON 5-6	MOM. 2-3 ON 5-6						
MS27785-21	1-2 ON 4-5	OFF	2-3 ON 5-6	7	7.5	7.5	4	7.5	7.5
MS27785-22	OFF	NONE	2-3 ON 5-6						
MS27785-23	1-2 ON 4-5	NONE	2-3 ON 5-6						
MS27785-24	NONE	OFF	2-3 ON 5-6						
MS27785-25	NONE	MOM. OFF	2-3 ON 5-6	5	7.5	7.5	2	7.5	7.5
MS27785-26	MOM 1-2 ON 4-5	NONE	2-3 ON 5-6						
MS27785-27	MOM. 1-2 ON 4-5	OFF	MOM. 2-3 ON 5-6						
MS27785-28	MOM. 1-2 ON 4-5	OFF	NONE						
MS27785-29	MOM OFF	NONE	2-3 ON 5-6						
MS27785-30	MOM. 1-2 ON 4-5	NONE	OFF						
MS27785-31	MOM. 1-2 ON 4-5	OFF	2-3 ON 5-6						
MS27785-32	NONE	1-2 ON 4-5	MOM. 2-3 ON 5-6						
MS27785-33	NONE	1-2 ON 4-5	2-3 ON 5-6	7	7.5	7.5	4	7.5	7.5

Test Requirements

Test shall be performed in accordance with MIL-DTL-3950 except

a) During all tests. Switches shall be fully wired with appropriate wire and terminal contacts.
b) Contact voltage drop- the contact drop with two terminals and the switch contact in series shall not exceed 8 millivolts measured from one wire contact through the contacts to the other wire contacts.

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Referenced documents
MIL-DTL-3950

The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA – CC

Preparing activity
DLA – CC

(Project 5930-2016-014)

Review activities
Army – AR, AV, MI
Navy – EC, MC
Air Force – 99

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